

Flux-CBH Cosmology: A Central-Black-Hole Entropy Reservoir Model for Late-Time Expansion and Structure Growth

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Abstract

The standard Λ CDM cosmological model remains highly successful, yet it faces persistent late-time tensions in the inferred Hubble constant H_0 and the amplitude of matter clustering, commonly parameterized by S_8 . In this work we introduce and test a phenomenological Flux-CBH cosmology in which the late-time dark sector is modeled as a dynamic response to an accumulated central-black-hole entropy reservoir. The framework is motivated by the Bekenstein-Hawking entropy scaling $S_{\text{BH}} \propto M_{\text{BH}}^2$, which implies that hierarchical black-hole growth and mergers act as a strongly nonlinear entropy-storage channel. We construct a redshift-dependent Flux response from a Schechter-integrated central black-hole entropy history and couple it to the background expansion through a dynamic compression exponent $p_{\text{eff}}(z)$. The resulting expansion history is tested against the CMB acoustic angle, supernova distance-shape residuals, BAO distance ratios, linear growth, and first-pass redshift-space-distortion data.

We identify two branches. An aggressive high- H_0 branch reaches $H_0 \simeq 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$, preserves the CMB acoustic angle at diagnostic accuracy, matches the supernova distance-redshift shape, and lowers S_8 to ~ 0.749 , but fails the absolute BAO distance ladder. A BAO-compatible branch settles at $H_0 \simeq 71 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m \simeq 0.288$, preserves the CMB acoustic angle, matches supernova shape residuals at the $\sim 0.014 \text{ mag}$ level, recovers a BAO fit comparable to Λ CDM, and lowers S_8 to ~ 0.800 while maintaining an RSD fit statistically equivalent to the Planck Λ CDM baseline. These results suggest that a late-time black-hole entropy reservoir can partially relax both the H_0 and S_8 tensions. A full $H_0 \simeq 73$ solution appears to require an additional early-sector or sound-horizon modification.

1 Introduction

The concordance Λ CDM model provides a compact and successful description of the cosmic microwave background, large-scale structure, baryon acoustic oscillations, and the late-time accelerated expansion of the Universe. Nevertheless, several observational tensions have become increasingly difficult to dismiss as mere statistical fluctuations. The most widely discussed are the Hubble tension, in which late-time distance-ladder measurements prefer a larger H_0 than CMB-inferred Λ CDM fits, and the S_8 tension, in which weak-lensing and low-redshift structure surveys prefer a lower clustering amplitude than the Planck Λ CDM prediction.

Many proposed resolutions modify either early-universe physics, late-time dark energy, gravitational growth, or neutrino/radiation content. However, a recurring difficulty is the inverse-distance-ladder constraint: models that raise H_0 at late times while preserving the early sound horizon often distort BAO distances. Conversely, models that alter the sound horizon can face pressure from CMB precision observables and structure formation.

This paper explores a phenomenological alternative inspired by black-hole thermodynamics. Supermassive and central black holes are not merely compact gravitating masses; they are the

dominant entropy reservoirs in the late Universe. Since Bekenstein-Hawking entropy scales quadratically with mass,

$$S_{\text{BH}} \propto M_{\text{BH}}^2, \quad (1)$$

hierarchical black-hole mergers create an entropy channel that grows faster than stellar mass or star-formation rate alone. This motivates the hypothesis that the late-time effective dark sector may track a compressed, redshift-dependent response to the accumulated central-black-hole entropy reservoir.

We call this phenomenological construction Flux-CBH cosmology. The goal is not to claim a complete fundamental theory, but to test whether a thermodynamic late-time reservoir can reproduce the observed background and growth phenomenology while relieving the modern tensions.

2 Physical Framework

2.1 Stellar current and black-hole entropy storage

A purely stellar source history peaks too early to resemble the late-time acceleration history. The Madau-Dickinson star-formation-rate density peaks near cosmic noon and declines toward the present. Its accumulated stellar-mass integral produces a smoother late-time reservoir, but still lacks the nonlinear amplification required to reproduce the observed dark-sector evolution.

Central black holes introduce the missing nonlinearity. For a Schwarzschild black hole,

$$\frac{S_{\text{BH}}}{k_B} \simeq 1.07 \times 10^{77} \left(\frac{M_{\text{BH}}}{M_{\odot}} \right)^2. \quad (2)$$

Thus, entropy density is not proportional to the mean black-hole mass alone. The high-mass tail dominates:

$$\langle M_{\text{BH}}^2 \rangle \neq \langle M_{\text{BH}} \rangle^2. \quad (3)$$

This makes black-hole mergers thermodynamically important. Two equal black holes of mass M have total entropy proportional to $2M^2$; after merger, the entropy scales as $(2M)^2 = 4M^2$, doubling the entropy at fixed total mass.

2.2 Schechter-integrated CBH entropy density

We model the central-black-hole entropy density as

$$S_{\text{CBH}}(z) = \int \Phi(M, z) S_{\text{BH}}(M) d \log M, \quad (4)$$

where $\Phi(M, z)$ is a redshift-dependent Schechter-like black-hole mass function. The precise mass-function parameterization is treated phenomenologically in the present diagnostic phase. The important feature is that the entropy density properly weights the massive end of the black-hole population through the quadratic Bekenstein-Hawking scaling.

A four-term Flux battery diagnostic is then constructed from stellar ignition, stellar mass, the CBH entropy reservoir, and the active CBH entropy-growth term:

$$F_{\text{raw}}(z) \sim A \dot{\rho}_{\star}(z) + B \rho_{\star}(z) + C S_{\text{CBH}}(z) + D \dot{S}_{\text{CBH}}(z), \quad (5)$$

where all terms are normalized for shape comparison. Non-negative fitting shows that CBH entropy terms dominate the late-time dark-sector shape, while stellar terms act as supporting channels.

2.3 Dynamic compression exponent

The metric does not respond linearly to raw black-hole entropy. A linear response is too aggressive. Instead, the diagnostic scans favor a compressed response,

$$F_{\text{metric}}(z) = F_{\text{raw}}(z)^{p_{\text{eff}}(z)}. \quad (6)$$

The effective compression exponent is redshift dependent:

$$p_{\text{eff}}(z) = p_{\text{high}} + \frac{p_0 - p_{\text{high}}}{1 + \exp[(z - z_t)/w]}. \quad (7)$$

The best-fit branches satisfy $p_{\text{high}} > p_0$, meaning that the metric response to the CBH entropy reservoir is stronger at earlier/post-cosmic-noon redshifts and softens toward the present epoch.

3 Background Cosmology

The Flux-CBH background is written as

$$E^2(z) = \Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_{\text{flux},0}F_{\text{metric}}(z), \quad (8)$$

with flat closure

$$\Omega_{\text{flux},0} = 1 - \Omega_m - \Omega_r. \quad (9)$$

The model is compared against a Planck-like Λ CDM baseline using several diagnostic observables.

3.1 CMB acoustic angle

The CMB acoustic angle is approximated by

$$\theta_{\star} = \frac{r_s(z_{\star})}{D_M(z_{\star})}, \quad (10)$$

where

$$r_s(z_{\star}) = \int_{z_{\star}}^{\infty} \frac{c_s(z)}{H(z)} dz, \quad (11)$$

and

$$D_M(z_{\star}) = \int_0^{z_{\star}} \frac{c}{H(z)} dz. \quad (12)$$

The present diagnostic uses acoustic-angle matching rather than a full Boltzmann likelihood.

3.2 Supernova shape residuals

Supernovae constrain the relative luminosity-distance shape rather than the absolute Hubble scale unless calibrated externally. The luminosity distance is

$$D_L(z) = (1+z) \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')}, \quad (13)$$

and the distance modulus is

$$\mu(z) = 5 \log_{10} \left(\frac{D_L(z)}{\text{Mpc}} \right) + 25. \quad (14)$$

We compare the intercept-marginalized residual

$$\Delta\mu_{\text{shape}}(z) = \Delta\mu(z) - \langle \Delta\mu \rangle_{z \in [0.01, 0.05]}. \quad (15)$$

3.3 BAO distances

BAO measurements compare distances to the drag-epoch sound horizon r_d . We compute

$$D_H(z) = \frac{c}{H(z)}, \quad (16)$$

$$D_V(z) = \left[z D_M^2(z) D_H(z) \right]^{1/3}, \quad (17)$$

and compare D_M/r_d , D_H/r_d , and D_V/r_d against a first-pass consensus BAO compilation.

4 Growth of Structure

The linear growth factor is integrated using

$$D'' + \left[2 + \frac{d \ln H}{dN} \right] D' - \frac{3}{2} \Omega_m(z) D = 0, \quad (18)$$

where $N = \ln a$, primes denote derivatives with respect to N , and

$$\Omega_m(z) = \frac{\Omega_{m,0}(1+z)^3}{E^2(z)}. \quad (19)$$

The first-pass calculation assumes

$$\mu_{\text{eff}} = 1, \quad (20)$$

so any change in growth comes from the background expansion history rather than an explicit modified-gravity growth coupling.

The growth rate is

$$f(z) = \frac{d \ln D}{d \ln a}, \quad (21)$$

and the observed RSD quantity is

$$f\sigma_8(z) = f(z)\sigma_8 \frac{D(z)}{D(0)}. \quad (22)$$

The weak-lensing summary statistic is

$$S_8 = \sigma_8 \sqrt{\frac{\Omega_m}{0.3}}. \quad (23)$$

5 Diagnostic Pipeline

The numerical pipeline proceeds in stages:

1. Integrate a Madau-Dickinson star-formation history and construct stellar-current and stellar-reservoir terms.
2. Build a Schechter-integrated CBH entropy density using the Bekenstein-Hawking scaling.
3. Fit a four-term Flux battery history and identify the CBH entropy reservoir as the dominant late-time component.
4. Introduce a compressed metric response $F_{\text{metric}} = F_{\text{raw}}^{\text{peff}}$.
5. Scan CMB acoustic angle and SN shape to identify high- H_0 branches.
6. Enforce BAO distances to identify BAO-compatible compromise branches.
7. Integrate the linear growth ODE and compare S_8 and RSD $f\sigma_8$ predictions.

6 Results

We identify two phenomenologically distinct Flux-CBH branches. The first, an aggressive high- H_0 branch, demonstrates that the dynamic CBH entropy reservoir can reproduce $H_0 \simeq 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$, preserve the early-universe CMB acoustic angle in the diagnostic calculation, match the late-time supernova distance-redshift shape, and suppress S_8 into the weak-lensing-preferred regime. However, this branch fails the absolute BAO distance ladder constraint. The second, a BAO-rescued branch, relaxes the Hubble shift to $H_0 \simeq 71.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$, while preserving the CMB acoustic angle, matching supernovae, recovering a BAO fit comparable to standard ΛCDM , and reducing S_8 to ~ 0.800 . These two branches bracket the present capability of the late-time-only Flux-CBH model.

6.1 Branch A: aggressive $H_0 \simeq 73$ solution

The first branch isolates the model's capacity to maximize Hubble-tension relief without altering the early-universe sound-horizon calibration. By allowing the physical matter density ω_m to float alongside the dynamic compression parameters p_0 and p_{high} , the model locates an island at

$$H_0 = 73.0 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad \Omega_m \simeq 0.263, \quad (24)$$

with

$$p_0 = 0.18, \quad p_{\text{high}} = 0.23. \quad (25)$$

This branch yields the following diagnostic results:

- CMB acoustic angle error: ~ 11 ppm.
- Maximum supernova shape residual: ~ 0.009 mag.
- Growth prediction: $S_8 \simeq 0.749$.
- RSD improvement relative to Planck ΛCDM : $\Delta\chi^2_{\text{RSD}} \simeq -2.83$.
- BAO fit: $\chi^2_{\text{BAO}} \simeq 40.6$, significantly worse than the ΛCDM baseline.

The BAO failure is localized primarily in the transverse comoving distances D_M/r_d at $z \sim 0.4\text{--}0.7$. The sound horizon remains nearly unchanged, so the failure is an inverse-distance-ladder effect rather than a sound-horizon instability.

6.2 Branch B: BAO-rescued compromise solution

To construct a BAO-compatible late-time background, Branch B enforces the absolute distance anchors of the BAO consensus compilation. Under this constraint, the Flux-CBH expansion history shifts into a compromise state:

$$H_0 = 71.0 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad \Omega_m \simeq 0.288, \quad (26)$$

with

$$p_0 = 0.19, \quad p_{\text{high}} = 0.25. \quad (27)$$

This branch gives:

- CMB acoustic angle error: ~ 17 ppm.
- Maximum supernova shape residual: ~ 0.014 mag.
- BAO fit: $\chi^2_{\text{BAO}} \simeq 17.7$, comparable to the ΛCDM diagnostic value $\chi^2 \simeq 16.8$.

- Growth prediction: $S_8 \simeq 0.800$, down from $S_8 \simeq 0.831$ in the Planck Λ CDM baseline.
- RSD fit: $\chi_{\text{RSD}}^2 = 9.98$, statistically equivalent to the Λ CDM diagnostic value $\chi_{\text{RSD}}^2 = 9.89$.

The contrast between Branch A and Branch B makes the inverse-distance-ladder constraint explicit. The CBH entropy reservoir can generate the late-time expansion history required for $H_0 \simeq 73$, and this same branch strongly suppresses S_8 . However, when the BAO absolute ruler is enforced without modifying the early sound horizon, the allowed solution moves toward $H_0 \simeq 71$. Thus, the present late-time-only model should be interpreted as a partial tension-relief mechanism, while the aggressive branch identifies the target behavior for a future extension including pre-recombination Flux dynamics.

6.3 Constraint summary

Table 1: Summary of the two Flux-CBH diagnostic branches.

Diagnostic	Branch A: aggressive	Branch B: BAO-rescued
H_0	73.0	71.0
Ω_m	~ 0.263	~ 0.288
p_0	0.18	0.19
p_{high}	0.23	0.25
CMB θ_*	Pass, ~ 11 ppm	Pass, ~ 17 ppm
SN shape	Pass, ~ 0.009 mag	Pass, ~ 0.014 mag
BAO distances	Fail, $\chi^2 \sim 40.6$	Pass, $\chi^2 \sim 17.7$
S_8	~ 0.749	~ 0.800
RSD $f\sigma_8$	Improved, $\Delta\chi^2 \sim -2.83$	Equivalent, $\Delta\chi^2 \sim +0.10$
Interpretation	full tension-relief potential	viable late-time compromise

7 Discussion

The diagnostic sequence shows that a black-hole entropy reservoir can reproduce several features usually attributed to a cosmological constant or a more general dark-energy sector. The model is most successful when the metric response to the CBH entropy history is compressed and redshift dependent. This supports the interpretation that spacetime does not respond linearly to raw black-hole entropy. Instead, the metric appears to read a screened or projected entropy channel.

The high- H_0 branch demonstrates the potential of the mechanism: it raises H_0 , lowers Ω_m , suppresses S_8 , and improves RSD consistency. However, it fails BAO, showing that a late-time-only modification cannot freely raise H_0 while leaving the sound horizon fixed.

The BAO-rescued branch is more conservative and more viable. It raises H_0 to ~ 71 , lowers S_8 to ~ 0.800 , and preserves BAO. This is not a complete resolution of the Hubble tension, but it is a simultaneous partial relaxation of both H_0 and S_8 .

The remaining theoretical implication is clear: a full $H_0 \simeq 73$ solution likely requires an early-sector extension. Possibilities include a pre-recombination Flux response, a modified drag-epoch sound horizon, or a coupling between the CBH entropy sector and earlier radiation/matter registration dynamics.

8 Limitations

This paper presents a diagnostic-level phenomenological model, not a full Boltzmann-code likelihood analysis. Several limitations remain:

1. The CBH entropy history is parameterized rather than derived from a fully observational black-hole mass-function catalog.
2. The CMB test uses the acoustic angle rather than a full temperature/polarization likelihood.
3. The supernova diagnostic uses shape residuals rather than the full Pantheon+ covariance matrix.
4. The BAO and RSD comparisons use first-pass consensus compilations and diagonal errors.
5. The growth calculation assumes $\mu_{\text{eff}} = 1$, leaving any explicit Flux modification to gravitational clustering for future work.
6. Transfer-function effects and matter-radiation equality consistency require a full Boltzmann implementation.

These limitations define the next stage of the research program rather than invalidating the present diagnostic result.

9 Conclusion

Flux-CBH cosmology provides a thermodynamically motivated late-time alternative to a strict cosmological constant. By treating central black holes as nonlinear entropy reservoirs and coupling their accumulated entropy history to the metric through a dynamic compression exponent, the model naturally produces late-time expansion histories capable of relaxing both the H_0 and S_8 tensions.

Two branches emerge. The aggressive branch reaches $H_0 \simeq 73$ and $S_8 \simeq 0.749$, but fails BAO. The BAO-rescued branch reaches $H_0 \simeq 71$, preserves CMB, SN, BAO, and RSD diagnostics, and lowers S_8 to ~ 0.800 . Thus, the late-time-only Flux-CBH model is best interpreted as a viable partial tension-relief mechanism. A complete high- H_0 solution likely requires an early-sector extension capable of modifying the sound-horizon calibration while preserving the successes of the late-time CBH entropy response.

A Accompanying Repository Scan Files

The following files are recommended for inclusion in the repository to make the diagnostic chain reproducible.

A.1 Core data product

- `flux_cbh_schechter_entropy.csv` - Main derived Flux-CBH entropy/metric-history table. Required by later scans.

A.2 Entropy-reservoir construction

- `flux_cbh_entropy_schechter.py` - Builds the Schechter-integrated central-black-hole entropy density and four-term Flux battery history.
- `delayed_memory_stellar_entropy_coupling.py` - Stellar-memory diagnostic motivating the CBH entropy reservoir.

A.3 CMB and Hubble-tension scans

- `flux_cbh_h0_tension_test_v2.py`
- `flux_cbh_cmb_rigid_grid_scan.py`
- `flux_cbh_matter_float_grid_scan.py`

A.4 Supernova and dynamic-response scans

- `scan8_fixed_dynamic_exponent.py`
- `scan9_throttle_fixed.py`
- `scan10_redshift_pivot.py`
- `scan10_archetype_compare.py`
- `scan11_dynamic_exponent_cmb_sn_joint.py`

A.5 Growth, RSD, and BAO scans

- `scan12_growth_ode.py`
- `scan13_fsigma8_rsd_fit.py`
- `scan14_bao_distance_fit.py`
- `scan15_joint_bao_rescue.py`
- `scan15_bao_rescue_results.csv`
- `scan16_bao_rescued_growth.py`

A.6 Final figures

- `scan11_best_joint_sn.png`
- `scan12_growth_history.png`
- `scan13_fsigma8_rsd_fit.png`
- `scan14_bao_distance_fit.png`
- `scan16_bao_rescued_growth.png`